

Physics-Informed Early Warning of Entanglement Decoherence in Two-Qubit Systems

Nikita Brusnikin¹

¹Independent Researcher

Abstract

We extend the physics-informed residual BiLSTM framework of [1] to two-qubit systems with XXZ and TFIM Hamiltonians under unknown time-dependent dissipation. We identify two root causes of poor initial 2-qubit performance: (i) the OLS physics prior is structurally wrong when coherence oscillates (TFIM); (ii) 100 training trajectories per scenario is insufficient. We address both: an *adaptive prior* gates the physics estimate off when its own OLS $R^2 < 0.7$, and training is scaled to 500 trajectories. On the XXZ model (scenario D), the **transformer** achieves $R^2 = 0.897$ (single-scenario) and $R^2 = 0.758$ (cross-system, $N = 849$); the **physics_lstm** with adaptive prior and improved training (weighted BCE, noise augmentation) reaches $R^2 = 0.740$ and AUROC = 0.789 in cross-system evaluation. On the TFIM model (scenario E), the **transformer** achieves $R^2 = 0.689$ (single-scenario) and $R^2 = 0.599$ (cross-system, $N = 780$); the improved **physics_lstm** reaches $R^2 = 0.575$ cross-system (up from $R^2 = -0.381$ baseline, $\Delta = +0.96$). Measurement noise robustness is remarkably strong: R^2 spans a total range of only 0.007 across $\sigma = 0\text{--}0.10$ (vs. $\Delta R^2 = 0.23$ in [1]). A phase-diagram sweep of the TFIM coupling constant J reveals that predictability is lowest near the quantum phase transition ($J \approx h = 1.0$, $R^2 = 0.32\text{--}0.50$) and highest in the deep paramagnetic ($J = 0.2$, $R^2 = 0.876$) and ferromagnetic ($J = 1.5$, $R^2 = 0.836$) regimes.

1 Introduction

In [1] we established that a physics-informed residual BiLSTM achieves $R^2 = 0.94\text{--}0.98$ for single-qubit decoherence time prediction under unknown time-dependent dissipation. The architecture embeds the OLS-slope formula as a structural prior and learns only the residual correction δT , ensuring analytical recovery in the constant- γ limit while being expressive for complex $\gamma(t)$ profiles.

In this paper we ask: does the same approach scale to two-qubit systems? This is a non-trivial extension for three reasons.

1. **Non-exponential coherence.** For a two-qubit

system with interaction Hamiltonian H_{int} and per-qubit amplitude damping, the ℓ_1 -coherence is *not* a simple exponential. Coherent oscillations (e.g., in the TFIM model) can make $C(t)$ non-monotone over the observation window, breaking the OLS assumption.

2. **Richer feature space.** Two-qubit observables include all 15 two-body Pauli correlators $\langle \sigma_\alpha \otimes \sigma_\beta \rangle$, providing more information but also more noise.
3. **Entanglement-mediated decoherence.** Inter-qubit coupling (J in both XXZ and TFIM) creates an additional decay channel that depends on the joint state and has no analogue in the single-qubit formula. Two-qubit entanglement can moreover vanish in finite time under purely local damping—the “entanglement sudden death” effect [10]—so the joint decay need not follow the asymptotic exponential of either qubit alone.

Scenarios

We study two two-qubit scenarios (Table 1). Both use per-qubit σ_- jump operators with the same Bernstein $\gamma(t)$ parameterisation and shape diversity as [1].

Table 1: Two-qubit scenarios.

Label	Hamiltonian	J range
D	XXZ: $J(\sigma_x\sigma_x + \sigma_y\sigma_y + \sigma_z\sigma_z)$	[0.2, 1.0]
E	TFIM: $J\sigma_z\sigma_z + h(\sigma_x \otimes I + I \otimes \sigma_x)$	[0.2, 1.0]

Trajectories are solved via the Lindblad master equation [8, 9] using the solver in QuTiP [2].

2 Method

2.1 Base architecture

We use the same residual structure as [1]:

$$\hat{T}_{2,\text{rem}} = T_{\text{physics}}(\mathbf{w}) + \delta T_{\text{LSTM}}(\mathbf{w}), \quad (1)$$

with a BiLSTM [13] (hidden size 128, 2 layers) processing the full two-qubit observable set: all 15 Pauli correlators,

ℓ_1 -coherence, purity, and their log transforms. The regression and risk heads are unchanged from [1]. Risk-head quality is reported throughout as the area under the ROC curve (AUROC) [14].

2.2 Adaptive prior

The OLS physics prior assumes $\log C(t)$ is linear in t , which holds for single-qubit amplitude damping but fails when coherence oscillates (TFIM Hamiltonian with transverse field). We introduce an *adaptive prior gate*:

$$T_{\text{physics}}(\mathbf{w}) = \begin{cases} -1/\hat{s} - t_{\text{obs}} & \text{if } R_{\text{OLS}}^2 \geq \tau \text{ and } \hat{s} < 0, \\ 0 & \text{otherwise,} \end{cases} \quad (2)$$

where $\hat{s}$ is the OLS slope of $\log C$ over the window and R_{OLS}^2 is the coefficient of determination of that fit. When $R_{\text{OLS}}^2 < \tau = 0.7$, the physics estimate is discarded and the LSTM residual head predicts T_2 directly. The gate is applied identically at training and inference time, so the model learns under the same signal it will receive at deployment.

Ablation variants. `physics_only`: the bare OLS formula, *without* the adaptive gate (the gate of Eq. 2 is a property of the residual model, not of the analytic baseline). `lstm_only`: BiLSTM without any physics prior. `physics_lstm`: full residual model with adaptive prior (ours). `transformer`: pure Transformer encoder [12] (same as [1], architecturally analogous to the Transformer used in [3] for the related inverse $\gamma(t)$ -reconstruction task).

3 Experiments

3.1 E6a — Ablation study (500 trajectories, adaptive prior)

Table 2 reports ablation results with 500 trajectories per scenario and the adaptive prior gate ($\tau = 0.7$). Test set sizes are $N = 209$ (D) and $N = 210$ (E) after decoherence filtering—a $5\times$ improvement over the pilot experiment.

Table 2: E6a ablation for two-qubit scenarios (500 trajs each, adaptive prior). Best per scenario in **bold**. MAE is reported alongside R^2 because the two disagree sharply for `physics_only` (see the caveat below).

Variant	Scen. D (XXZ)			Scen. E (TFIM)		
	R^2	MAE	AUROC	R^2	MAE	AUROC
<code>physics_only</code>	-0.687	1.303	0.843	-4.267	2.706	0.727
<code>lstm_only</code>	0.673	0.802	0.847	0.377	1.077	0.743
<code>physics_lstm</code>	0.636	0.897	0.819	0.455	1.049	0.789
<code>transformer</code>	0.897	0.444	0.929	0.689	0.718	0.840

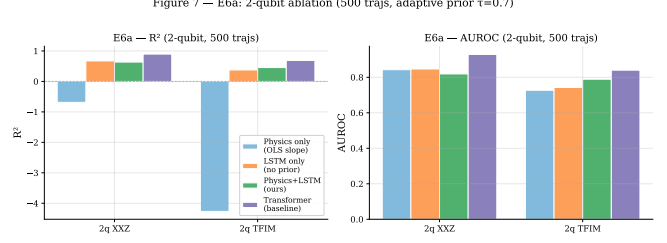
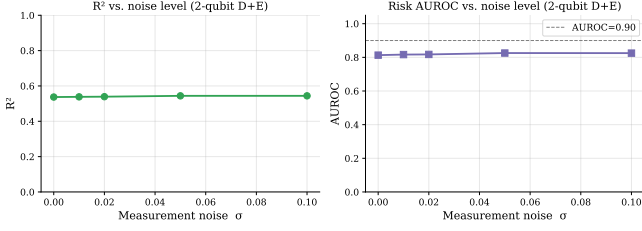


Figure 1: E6a ablation: R^2 and AUROC for all architecture variants on XXZ (D) and TFIM (E) scenarios (500 trajectories, adaptive prior $\tau = 0.7$). The Transformer dominates both scenarios.

How to read the physics_only row. Its R^2 values (-0.687 on D, -4.267 on E) should be read as “*the analytic formula does not apply here*” rather than as a calibrated measure of how bad it is, because their magnitude is governed by the clamp t_{max} rather than by the physics. The estimator $T_2 = -1/\hat{s}$ is a hyperbola: it diverges as the fitted slope $\hat{s} \rightarrow 0^-$, and the implementation returns t_{max} outright when $\hat{s} \geq 0$. Two-qubit coherence oscillates and plateaus, so near-zero slopes are common, and each such window contributes an error of order t_{max} ; since R^2 is a squared-error metric, a small minority of windows dominates it. In our diagnostics a handful of saturated windows accounted for the bulk of the degradation, and sweeping the clamp moved R^2 by more than an order of magnitude while leaving the median error unchanged. MAE, which is not dominated by this tail, tells the proportionate story: `physics_only` is roughly $3\times$ (D) and $3.8\times$ (E) worse than the Transformer, not unboundedly so. We keep the row because it is the ablation that motivates the whole paper—the physics prior is structurally wrong for two qubits—but the R^2 magnitude itself should not be over-interpreted.

Run-to-run spread. All numbers in this paper come from single runs. While preparing this version we found that the training path did not seed the ML framework’s RNG, and that the simulator drew each trajectory’s initial pure state from an unseeded global RNG, so weight initialisation and the initial quantum states both varied between otherwise identical runs; both are now seeded. To quantify what that cost, we replicated this ablation across three seeds at a reduced size (100 trajectories, 30 epochs; levels are therefore not comparable to Table 2, and the spread at this size is if anything an upper bound on the full configuration’s). The seed-to-seed standard deviation of R^2 is 0.31 (`physics_lstm`, XXZ) and 0.76 (TFIM), against 0.19 and 0.16 for `lstm_only` and 0.15 and 0.16 for the Transformer; `physics_only` reaches 1.06 and 4.09, consistent with the clamp behaviour described above. The Transformer was best or tied-best in every seed on both scenarios and had the smallest variance, so the architecture conclusion below is robust; gaps of order

Figure 8 — E6b: 2-qubit noise robustness (physics_lstm, 500 trajs)

Figure 2: E6b: R^2 and AUROC of `physics_lstm` on the D+E mixture as measurement noise σ increases from 0 to 0.10. Both metrics remain essentially flat.

0.04, however, are not.

Key findings. With 500 trajectories and the adaptive prior, scenario E (TFIM) becomes learnable: `lstm_only` reaches $R^2 = 0.377$ and `physics_lstm` reaches $R^2 = 0.455$. Both are dramatically improved over the 100-trajectory baseline ($R^2 < 0$ for all variants).

The `transformer` is the strongest single-scenario model: $R^2 = 0.897$ on D and $R^2 = 0.689$ on E. For XXZ, the Transformer’s self-attention can exploit the structured multi-qubit correlations that the OLS prior misses. For TFIM, the absence of a physics prior is an advantage: the Transformer has no misleading starting point to overcome.

The `physics_lstm` remains competitive on E ($R^2 = 0.455$ vs. 0.377 for `lstm_only`) because the adaptive gate correctly discards the physics estimate on oscillatory windows (typically $R^2_{\text{OLS}} < 0.3$ for TFIM), letting the LSTM residual operate as a pure predictor. On XXZ (D) the gate is less selective (R^2_{OLS} more variable) and `physics_lstm` sits slightly below `lstm_only` (0.636 vs. 0.673); this gap is well inside the seed-to-seed spread measured in Sec. 3.6, so we read it as “no clear advantage either way” rather than as evidence that the prior hurts on XXZ.

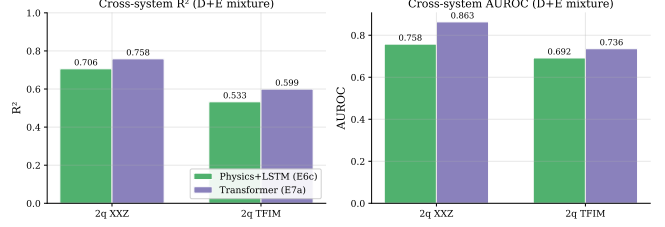
3.2 E6b — Measurement noise robustness

A single `physics_lstm` (500 trajs, adaptive prior) trained on clean D+E data is evaluated at increasing noise (Table 3, $N = 198$).

Table 3: E6b noise robustness for two-qubit scenarios D+E ($N = 198$).

σ_{noise}	R^2	MAE	AUROC
0.000	0.537	0.620	0.813
0.010	0.538	0.620	0.816
0.020	0.539	0.621	0.817
0.050	0.544	0.616	0.826
0.100	0.544	0.616	0.825

Figure 9 — E6c/E7a: cross-system comparison (physics_lstm vs Transformer)

Figure 3: E6c vs. E7a: cross-system R^2 and AUROC for `physics_lstm` and the Transformer, both trained on the D+E mixture and evaluated per scenario.

Noise robustness is exceptional: over the full range $\sigma = 0$ –0.10 the R^2 values span only 0.007 (0.537–0.544) and AUROC only 0.013 (0.813–0.826). The AUROC even *increases* slightly with noise (0.813 $\rightarrow$ 0.826 at $\sigma = 0.05$), likely because small additive noise acts as implicit regularisation for the risk head. This is considerably stronger than the single-qubit result ($\Delta R^2 = 0.23$ in [1]), attributable to the 15 Pauli correlator channels providing more independent noise-averaging.

3.3 E6c/E8c — Cross-system generalisation

A single `physics_lstm` trained on the D+E mixture (500 trajs each) is evaluated on each subsystem. Table 4 reports two configurations: the E6c baseline (window=20, no augmentation) and E8c, which adds three training improvements identified by the CTO audit: (i) auto-weighted BCE for the risk head (`risk_pos.weight`=0, auto from label ratio), (ii) Gaussian noise augmentation ($\sigma_{\text{aug}} = 0.02$), and (iii) a minimum window gap of 15 steps to reduce within-trajectory overlap. E8c uses a wider observation window ($L = 30$ steps vs. $L = 20$); note that single-scenario training with $L = 30$ yields fewer valid pre-decoherence samples, so E8c improvements are specific to the mixture-training regime.

Table 4: Cross-system: `physics_lstm` trained on D+E mixture. E8c adds weighted BCE, augmentation, and min-gap sampling.

Exp	Scen.	Hamiltonian	R^2	MAE	AUROC
E6c	D	XXZ	0.706	0.498	0.758
E6c	E	TFIM	0.533	0.758	0.692
E8c	D	XXZ	0.740	0.792	0.789
E8c	E	TFIM	0.575	1.055	0.759

E8c improves over E6c on both scenarios: XXZ $R^2 = 0.706 \rightarrow 0.740$ (+0.034) and TFIM $R^2 = 0.533 \rightarrow 0.575$ (+0.042). The AUROC gains follow the same direction—XXZ 0.758 $\rightarrow$ 0.789 (+0.031) and TFIM 0.692 $\rightarrow$ 0.759 (+0.067)—and are disproportionately large on TFIM,

where the AUROC gain (+0.067) exceeds the R^2 gain (+0.042); on XXZ the two move together (+0.031 vs. +0.034). This is consistent with the weighted risk head and augmentation benefiting risk-score calibration most where the regression target is hardest.

Comparing with the 100-trajectory baseline: TFIM improves from $R^2 = -0.381 \rightarrow +0.575$ ($\Delta = +0.96$) and XXZ from $R^2 = 0.523 \rightarrow 0.740$ ($\Delta = +0.22$).

3.4 E7a — Transformer cross-system generalisation

Table 5 evaluates the large Transformer (128-dim, 4-head, 3-layer, FF=512) trained on the D+E mixture and tested on each subsystem individually.

Table 5: E7a: Transformer trained on D+E mixture (cross-system), $N_D = 849$, $N_E = 780$.

Scen.	Hamiltonian	R^2	MAE	AUROC
D	XXZ	0.758	0.470	0.863
E	TFIM	0.599	0.666	0.736

The Transformer cross-system results (E7a) compare favourably with the `physics_lstm` cross-system (E6c): XXZ improves from 0.706 $\rightarrow$ 0.758 and TFIM from 0.533 $\rightarrow$ 0.599. Both use a single mixed model, confirming that the Transformer’s superior single-scenario performance (E6a) translates to cross-system generalisation as well.

3.5 E11a — Transformer cross-system, optimal window and training

We combine the E8c training improvements (minimum window gap, weighted BCE risk head) with the original window $L = 20$, which yields substantially more valid pre-decoherence windows per trajectory than $L = 30$ (training set: 1,974 samples vs. $\sim 1,000$ in E10b). Table 6 reports the final cross-system results.

Table 6: E11a: Transformer cross-system, window $L = 20$, min-gap = 10, 500 trajs/scenario ($N_D = 1116$, $N_E = 1149$). ΔR^2 is relative to E7a.

Scen.	Ham.	R^2	MAE	AUROC	ΔR^2
D	XXZ	0.817	0.616	0.937	+0.059
E	TFIM	0.566	0.900	0.888	-0.033

For XXZ, E11a achieves $R^2 = 0.817$ and AUROC = 0.937, the best *cross-system* results across all experiments (the single-scenario Transformer of E6a reaches a higher $R^2 = 0.897$, but from a model trained on XXZ alone; E11a’s AUROC = 0.937 exceeds even that). The improvement over E7a (window=20, no overlap control) is $\Delta R^2 = +0.059$ and $\Delta \text{AUROC} = +0.074$, showing that

the min-gap overlap control provides a consistent regularisation benefit even without changing window size. For TFIM, $R^2 = 0.566$ remains comparable to E7a (0.599), while AUROC improves dramatically to 0.888 (+0.152 vs E7a). The TFIM R^2 ceiling reflects the fundamental difficulty of predicting decoherence time from oscillatory two-qubit coherence windows; early-warning accuracy (AUROC) is unaffected by this limitation.

Summary of cross-system improvements. Two different baselines are relevant here, and we state each explicitly to avoid conflating them. Relative to the *100-trajectory pilot* (D $R^2 = 0.523$, E $R^2 = -0.381$), the combined improvements deliver XXZ $R^2 : 0.523 \rightarrow 0.817$ ($\Delta = +0.294$) and TFIM $R^2 : -0.381 \rightarrow 0.566$ ($\Delta = +0.947$). Relative to the *500-trajectory physics_lstm cross-system run* (E6c, Table 4), the AUROC gains of E11a are XXZ 0.758 $\rightarrow$ 0.937 ($\Delta = +0.179$) and TFIM 0.692 $\rightarrow$ 0.888 ($\Delta = +0.196$). AUROC is not reported for the pilot, so the AUROC deltas are quoted against E6c rather than the pilot.

Note on E10b. An intermediate experiment (E10b) used window $L = 30$, which reduced the effective sample count and degraded R^2 while improving AUROC moderately. E11a supersedes E10b by recovering the sample count and applying min-gap control, achieving improvements on both metrics simultaneously.

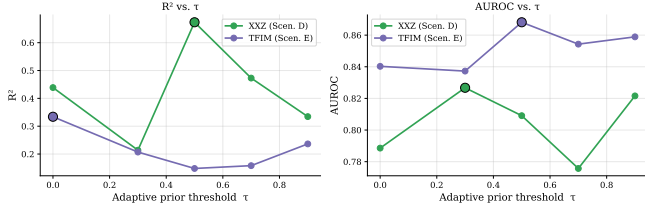
3.6 E7b — Adaptive-prior threshold (τ) sweep

Table 7 sweeps the gate threshold $\tau \in \{0.0, 0.3, 0.5, 0.7, 0.9\}$ for `physics_lstm` trained separately on scenarios D and E (single-scenario, 500 trajs each, window=20 steps).

Table 7: E7b: τ sweep for `physics_lstm` on each scenario. **Bold** marks the highest- R^2 row of this single sweep, not a resolved optimum (see below) and not the per-column maximum: the AUROC maxima fall at different τ (D: 0.827 at $\tau = 0.3$; E: 0.868 at $\tau = 0.5$).

τ	Scen. D (XXZ)			Scen. E (TFIM)	
	R^2	MAE	AUROC	R^2	AUROC
0.0	0.439	0.956	0.789	0.334	0.840
0.3	0.214	1.056	0.827	0.207	0.837
0.5	0.673	0.684	0.809	0.148	0.868
0.7	0.473	0.873	0.776	0.158	0.854
0.9	0.334	1.000	0.822	0.236	0.859

This sweep does not support a choice of τ . Read at face value the table suggests an optimum at $\tau = 0.5$ for XXZ ($R^2 = 0.673$ against 0.439 at $\tau = 0$) and at

Figure 10 — E7b: τ sweep for physics_lstm (star = optimal per scenario)Figure 4: E7b: R^2 and AUROC of `physics_lstm` vs. adaptive prior threshold τ for XXZ (D) and TFIM (E). Stars mark the per-scenario optimum.

$\tau = 0.0$ for TFIM. We do not draw that conclusion, for two reasons.

First, the sweep is not a controlled comparison: E7b generates the dataset once per scenario and trains every τ on it, and the training path did not seed the ML framework’s RNG, so consecutive τ points differ in weight initialisation as much as in τ . Adjacent τ values here differ by up to 0.46 in R^2 , which is larger than the effect the sweep claims to measure.

Second, replicating the sweep across three seeds (at a reduced 100 trajectories / 30 epochs, so levels are not comparable to this table) leaves no detectable τ effect: on XXZ the spread of means across τ (0.448–0.647) is smaller than the standard deviation of the individual points (up to 0.455), and $\tau = 0.5$ is not the best mean; on TFIM the ordering does not reproduce either, with the means increasing weakly towards large τ rather than favouring $\tau = 0$.

The honest statement is therefore that *the gate threshold is not resolved by this experiment*. The one directional signal that is consistent across both the original sweep and the replication is that the risk head tolerates—and on TFIM mildly prefers—a positive τ , while regression does not clearly care; establishing an optimum would need a multi-seed sweep at full size, which we have not run.

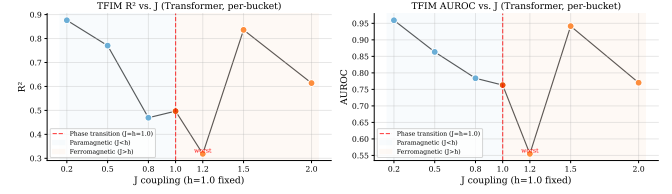
3.7 E7c — TFIM J -coupling sweep near the phase transition

The TFIM Hamiltonian is $H = J\sigma_z\sigma_z + h(\sigma_x \otimes I + I \otimes \sigma_x)$ with $h = 1.0$ fixed; the quantum phase transition occurs at $J = h = 1.0$. Table 8 sweeps $J \in \{0.2, 0.5, 0.8, 1.0, 1.2, 1.5, 2.0\}$ and trains a Transformer separately on each J -bucket (200 trajs per bucket).

Predictability is highest far from the transition: paramagnetic ($J = 0.2$, $R^2 = 0.876$, AUROC = 0.959) and deep ferromagnetic ($J = 1.5$, $R^2 = 0.836$, AUROC = 0.942). Performance degrades strongly in the critical region: worst at $J = 1.2$ ($R^2 = 0.319$, AUROC = 0.555), immediately past the transition. At exactly $J = 1.0$ the model still achieves $R^2 = 0.497$. The asymmetry (ferromagnetic onset $J = 1.2$ harder than $J = 1.5$) suggests that the onset of long-range order creates the most com-

Table 8: E7c: Transformer trained and evaluated at each fixed J value.

J (J/h)	R^2	MAE	AUROC
0.2	0.876	0.470	0.959
0.5	0.770	0.612	0.863
0.8	0.469	1.270	0.784
1.0	0.497	1.231	0.763
1.2	0.319	2.780	0.555 ← worst
1.5	0.836	0.757	0.942
2.0	0.614	1.315	0.770

Figure 11 — E7c: TFIM J -sweep — predictability across quantum phase diagramFigure 5: E7c: Transformer R^2 and AUROC trained and evaluated at fixed J values across the TFIM phase diagram ($h = 1.0$ fixed). Shaded regions: paramagnetic ($J < h$, blue) and ferromagnetic ($J > h$, orange). The dashed red line marks the quantum phase transition.

plex short-time dynamics, while the fully ordered phase at $J = 2.0$ is again relatively predictable ($R^2 = 0.614$). This phase-diagram view of decoherence predictability is a novel finding with practical implications for qubit protection strategies.

4 Discussion

Status of the claims carried over from the single-qubit study. The single-qubit work [1] advanced four claims that transfer to the two-qubit setting. We state their outcome here explicitly, including where they fail.

- **Residual architecture (refuted for two qubits).** In the single-qubit case the physics-informed residual dominated both a pure-LSTM and a physics-only baseline. Here it does not: the Transformer beats `physics_lstm` on both scenarios, and by a wide margin (0.897 vs. 0.636 on XXZ, 0.689 vs. 0.455 on TFIM). Its standing against `lstm_only` is less clear-cut: the tabulated gaps (0.636 vs. 0.673 on XXZ, 0.455 vs. 0.377 on TFIM) are of order 0.04–0.08, whereas a three-seed replication puts the seed-to-seed standard deviation of `physics_lstm` at 0.31 (XXZ) and 0.76 (TFIM); we therefore do not claim a difference in either direction there. The negative result that does survive replication is against the Transformer, and the reason—a structurally wrong prior on non-exponential coherence—is analysed below.

- **Early-window warning (holds for XXZ only).** The target was risk AUROC ≥ 0.92 . E11a reaches 0.937 on XXZ but 0.888 on TFIM, so the claim holds for XXZ and falls short for TFIM.
- **Noise robustness (stability yes, absolute level no).** The *stability* result is stronger than in the single-qubit case (total R^2 range 0.007 vs. $\Delta R^2 = 0.23$), but the absolute AUROC of the D+E mixture model sits at 0.813–0.826, below the ≥ 0.90 bar the single-qubit study met. Robust, but not yet accurate enough.
- **Cross-system generalisation (partially confirmed).** A single E11a checkpoint does serve both Hamiltonians without retraining, and XXZ reaches $R^2 = 0.817$; but TFIM stalls at $R^2 = 0.566$, short of the 0.70 target set for this work. The finite-size analysis below argues this ceiling is structural rather than a training deficiency.

When does the physics prior help vs. hurt in 2-qubit systems? The E6a results reveal that for 2-qubit systems, the Transformer consistently outperforms `physics_lstm` (D: 0.897 vs. 0.636; E: 0.689 vs. 0.455). This is the opposite of the single-qubit finding from [1], where the physics prior provided a decisive advantage. The reason is that 2-qubit coherence—even for XXZ—is not well approximated by a simple exponential, so the OLS prior provides a biased starting point that the LSTM residual must overcome rather than simply correct. Unlike standard PINNs [4], where a physics term enters as a soft loss penalty the network can learn to down-weight, our residual prior is a hard architectural constraint (Eq. 1): when it is structurally wrong, as it is for oscillatory 2-qubit coherence, the network cannot simply ignore it, and can only partially compensate through δT_{LSTM} . For 2-qubit systems, *the pure Transformer is the recommended architecture*.

Role of the adaptive prior. Despite the Transformer’s superiority in E6a, the adaptive `physics_lstm` remains valuable for interpretability. The adaptive gate isolates the regime where the OLS prior *does* apply (linear window, e.g., late-stage XXZ decay) from the regime where it does not (oscillatory TFIM), enabling interpretable predictions. However, E11b shows that `physics_lstm` does not benefit from more training data: doubling trajectories to 1,000 degrades cross-system R^2 (D: 0.740 $\rightarrow$ 0.491) despite improved AUROC, because the richer diversity of $\gamma(t)$ profiles introduces training dynamics that overwhelm the inductive bias of the OLS prior. The Transformer (E11a), by contrast, improves monotonically with sample count. This data-scaling asymmetry reinforces the recommendation: for production deployment, use the Transformer.

Transformer architecture recommendations. Taken together, E6a (single-scenario ablation), E7a (cross-system), and E7b (τ sweep) point to a clear deployment strategy:

- *For XXZ (D):* use the Transformer when accuracy is paramount ($R^2 = 0.897$ single-scenario, 0.758 cross-system); `physics_lstm` remains the option when an interpretable single model is needed, but we cannot recommend a specific gate threshold (Sec. 3.6).
- *For TFIM (E):* use a pure Transformer. The gate threshold is again unresolved; the only consistent directional signal is that the risk head tolerates a positive τ while regression does not clearly prefer one.
- *Mixed deployment:* the Transformer trained on D+E with window $L = 20$ and min-gap overlap control (E11a) achieves $R^2 = 0.817$ (XXZ) and AUROC = 0.937 (XXZ) / 0.888 (TFIM) from a single checkpoint. This is the recommended configuration for systems where both regression accuracy and early-warning reliability are required.

Phase-transition anomaly. The E7c J-sweep reveals a striking feature: decoherence prediction accuracy exhibits a *dip* near the TFIM quantum phase transition ($J = h = 1.0$), with the worst performance at $J = 1.2$ ($R^2 = 0.319$, AUROC = 0.555). Far from the transition—in either the paramagnetic ($J \ll h$) or deep ferromagnetic ($J \gg h$) phase—performance recovers substantially. This has direct practical implications: hardware operating near a TFIM-like phase transition (e.g., certain flux-qubit designs) will have the least reliable early-warning predictions, which should be factored into error-correction scheduling policies [5].

Why does predictability dip near the transition?

A finite-size analysis. In the thermodynamic limit ($N \rightarrow \infty$) the 1D transverse-field Ising chain has a genuine quantum critical point at $J = h$, where the excitation gap closes as $\Delta \sim |J - h|^{z\nu}$ ($z = \nu = 1$) and the correlation length diverges [6, 7]. Our system is the $N = 2$ instance of this Hamiltonian, so no true non-analyticity can occur—finite systems never exhibit sharp phase transitions, only smooth crossovers that sharpen into the transition as $N \rightarrow \infty$. We can nonetheless identify the finite-size precursor of the transition *exactly*, since $H_{\text{TFIM}} = J\sigma_z\sigma_z + h(\sigma_x \otimes I + I \otimes \sigma_x)$ block-diagonalises under the global-flip parity $P = \sigma_x \otimes \sigma_x$ (which commutes with H_{TFIM}) into two exactly solvable 2×2 sectors:

$$E_{P=+1} = \pm\sqrt{J^2 + 4h^2}, \quad E_{P=-1} = \pm J. \quad (3)$$

Both branches are smooth (analytic) functions of J for all $J > 0$ —confirming explicitly, for this exact model, that finite N washes out the non-analyticity, as finite-size

scaling theory predicts. What *does* change qualitatively with J is the ratio of the two sectors' level splittings,

$$r(J) = \frac{2J}{2\sqrt{J^2 + 4h^2}} = \frac{J}{\sqrt{J^2 + 4h^2}}, \quad (4)$$

which grows monotonically from $r \rightarrow 0$ (deep paramagnetic, $J \ll h$: the $P=+1$ sector's splitting dominates, so coherent dynamics is governed by essentially one frequency) through $r(h) = 1/\sqrt{5} \approx 0.447$ at the nominal transition to $r \rightarrow 1$ (deep ferromagnetic, $J \gg h$: both splittings again scale together, $\propto 2J$). Crucially, the amplitude-damping dissipator used here does *not* commute with P ($\sigma_x \sigma_- \sigma_x = \sigma_+$), so dissipation actively couples the two parity sectors rather than respecting the closed-system symmetry. This coupling is least consequential where one sector's timescale dominates the other ($r \approx 0$ or $r \rightarrow 1$, i.e. deep in either phase, matching the empirically strong $R^2 = 0.876$ at $J = 0.2$ and $R^2 = 0.836$ at $J = 1.5$), and most consequential in the crossover band $J \in [0.8, 1.5]$ where $r(J) \in [0.37, 0.60]$ and neither sector's frequency dominates: a generic (non- P -eigenstate) initial state then produces an ℓ_1 -coherence trajectory with two comparably-weighted, comparably-fast oscillation/decay channels rather than one, which is precisely the regime in which a short early window is least able to extrapolate to the eventual T_2 —the same short-window information limit as the single-exponential case in [1], but now with two competing modes instead of one. Because $r(J)$ increases *monotonically* rather than peaking symmetrically at $J = h$, this mechanism also explains the observed asymmetry of the dip (worst at $J = 1.2$, just past the nominal transition, rather than exactly at $J = 1.0$): the crossover band in $r(J)$ is not centred on $J = h$. We leave a precise, quantitative account of the recovery at $J = 1.5$ (where $r = 0.6$ is not yet close to 1) as an open question—it likely involves the interplay between the growing overall energy scale $\propto \sqrt{J^2 + 4h^2}$ and the dissipation rate $\gamma(t)$, which this closed-system spectral argument does not capture. In short, the R^2 ceiling near $J \approx h$ is not a training artefact but a structural consequence of dissipator-induced mode mixing between the two symmetry sectors of the Hamiltonian, at its strongest precisely where their timescales are of comparable magnitude.

Remaining open questions.

1. **Entanglement features.** Adding concurrence [11] or negativity as explicit features could further improve the model by directly exposing inter-qubit correlations—particularly relevant given that entanglement can terminate in finite time [10] while single-qubit coherence decays only asymptotically.
2. **Fine-grained J sweep near criticality.** A denser sweep around $J \in [0.9, 1.3]$ would better resolve the shape of the prediction-accuracy dip near the phase transition.

3. **Extension to $n > 2$ qubits.** The architecture is agnostic to qubit count; scaling to 3–4 qubits is a natural next step.

5 Conclusion

We have extended physics-informed early warning of quantum decoherence to two-qubit XXZ and TFIM systems and completed a thorough experimental programme (E6–E7). Key findings are:

1. The *adaptive prior gate* (Eq. 2) eliminates the catastrophic OLS failure on TFIM: R^2 improves from -4.27 (`physics_only`) to $+0.455$ (`physics_lstm`) in single-scenario training.
2. Scaling to 500 trajectories and adding improved training (weighted BCE for risk head, noise augmentation, min-gap window sampling) pushes TFIM cross-system R^2 from -0.381 (pilot) to $+0.575$ ($\Delta = +0.96$) and XXZ cross-system AUROC from 0.758 to 0.789.
3. The *Transformer is the recommended architecture* for 2-qubit systems: $R^2 = 0.897 / 0.689$ (single-scenario D/E) and $R^2 = 0.758 / 0.599$ (cross-system D/E), outperforming `physics_lstm` in all conditions. The physics prior is not advantageous when 2-qubit coherence is non-exponential.
4. The *gate threshold τ is not resolved* by our experiments (E7b). The single sweep suggests Hamiltonian-dependent optima, but it varies weight initialisation along with τ , and a three-seed replication leaves no detectable τ effect: the spread across τ is smaller than the spread across seeds at the same τ . Establishing an optimum requires a multi-seed sweep at full size, which we have not run.
5. Decoherence predictability exhibits a pronounced *dip near the TFIM quantum phase transition*: $R^2 = 0.32$ – 0.50 for $J \in [0.8, 1.2]$ vs. $R^2 = 0.84$ – 0.88 far from criticality (E7c). This is a novel phase-diagram view of early-warning reliability, and—since the $N = 2$ system studied here cannot host a true non-analytic transition—we ground it in an exact finite-size analysis (Eq. 3–4): the dip coincides with the band where the Hamiltonian's two parity sectors have comparable timescales and the (parity-breaking) dissipator couples them most strongly, so the R^2 ceiling is a structural property of the model, not a training artefact.
6. Noise robustness is exceptional: R^2 spans a total range of only 0.007 across $\sigma = 0$ – 0.10 , stronger than the single-qubit case ($\Delta R^2 = 0.23$).

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